

UE23CS352A: MACHINE LEARNING

Week 14: CNN Image Classification

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1. Introduction

The objective of this lab was to build a complete end-to-end image classification pipeline using a Convolutional Neural Network (CNN) to recognize hand gestures for Rock, Paper, and Scissors. The task involved loading and preparing a real-world dataset, constructing a CNN from scratch, training it using PyTorch, evaluating its accuracy, and finally using the trained model to predict the outcome of a Rock–Paper–Scissors match. This lab helped demonstrate how deep learning models can process visual information and make meaningful predictions based on learned patterns.

2. Model Architecture

The CNN architecture designed for this lab consists of three convolutional blocks. Each block contains a 3×3 convolution layer followed by ReLU activation and a MaxPooling layer with a stride of 2. The number of channels gradually increases from 16 to 32 and finally 64, allowing the network to learn progressively richer visual features.

The MaxPooling layers reduce the spatial dimensions from 128×128 down to 16×16 after three pooling operations. After feature extraction, the network uses a fully-connected classifier consisting of a Flatten layer, a Linear layer that maps the 16,384 features to 256 units, a ReLU activation, a Dropout layer with a probability of 0.3, and a final Linear layer that outputs the three class logits (rock, paper, scissors).

This combination of convolutional and dense layers allows the model to learn both spatial and semantic patterns from images.

3. Training and Performance

The model was trained using the Adam optimizer with a learning rate of 0.001 and CrossEntropyLoss as the loss function.

The training was run for 10 epochs, and the loss steadily decreased from 0.82 to just 0.01, showing strong convergence.

After training, the model was evaluated on a test set of 438 images and achieved a high accuracy of **97.49%**, demonstrating that the CNN learned the dataset patterns very effectively. The model was also successfully tested on random images, where it correctly predicted each class and determined the winner in a simulated Rock-Paper-Scissors match.

4. Conclusion and Analysis

Overall, the model performed extremely well, achieving nearly 97.5% test accuracy with a relatively simple CNN architecture. The training process was smooth, and the model showed strong generalization across the dataset.

One minor challenge was ensuring consistent image preprocessing, as the transform pipeline must match exactly during both training and inference.

To further improve the model, adding data augmentation (like rotation, flipping, or random brightness changes) could make the model more robust to variations in hand shapes and backgrounds. Additionally, experimenting with deeper architectures or transfer learning (e.g., using MobileNet or ResNet) could push the accuracy even higher.